STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 3: Motion in a Plane

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Is displacement scalar or vector? [1]
2. x-component at angle theta? [1]
3. Horizontal acceleration of an ideal projectile? [1]
4. Direction of centripetal acceleration? [1]
5. Magnitude of perpendicular components 3 and 4? [1]
6. Relation between v and omega? [1]
7. Ideal maximum-range angle at equal heights? [1]
8. Vertical velocity at projectile peak? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. Resolve 10 N at 60. [2]
10. Displacement after a full circular lap? [2]
11. v=12 m/s, r=6 m: a_c? [2]
12. Horizontal launch from 20 m, g=10: flight time? [2]
13. Perpendicular vectors 6 and 8: resultant? [2]
14. 120 rpm in rad/s? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. Projectile at 20 m/s, 30, g=10: T,H? [3]
16. Forces 8 N east and 6 N north: resultant? [3]
17. Cyclist at 10 m/s, radius 25 m: a_c? [3]
18. 25 m/s at 37, sin=.6, cos=.8, g=10: time/range? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Projectile 20 m/s at 30: time, height, range? [4]
20. 5 kg at 4 m/s in radius 2 m: inward force and work? [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations worth knowing

- Vector components: A_x=A cos(theta); A_y=A sin(theta).
- Perpendicular resultant magnitude: R = sqrt(A_x^2 + A_y^2); direction theta = tan^-1(A_y/A_x).
- Angular speed: omega = v/r; centripetal acceleration a_c = v^2/r = r omega^2.
- Projectile components: u_x=u cos(theta); u_y=u sin(theta).
- For launch and landing at the same height: T=2u sin(theta)/g; H=u^2 sin^2(theta)/(2g); R=u^2 sin(2theta)/g.
- Vertical drop from rest: h=(1/2)gt^2; horizontal launch flight time t=sqrt(2h/g); horizontal range = u_x t.
- Angular conversion: 1 revolution = 2pi radians; omega=2pi f.

Use each equation only when its assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Solutions are grouped here after the question paper and formula reference.

Question 1

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Vector.
Final answer: Vector.

Question 2

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Acostheta.
Final answer: Acostheta.

Question 3

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Zero.
Final answer: Zero.

Question 4

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Inward.
Final answer: Inward.

Question 5

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: 5.
Final answer: 5.

Question 6

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: v=romega.
Final answer: v=romega.

Question 7

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: 45.
Final answer: 45.

Question 8

Principle / definition: Recall the relevant definition or equation from the formula reference.
Apply it to this question: Zero.
Final answer: Zero.

Question 9

Working: Resolve the 10 N vector: A_x=10 cos60 deg=10(0.5)=5 N; A_y=10 sin60 deg=10(0.866)=8.66 N.

Final answer: A_x=5 N; A=8.66 N.

Question 10

Working: After one complete lap, final position equals initial position. Displacement=final position - initial position=0.

Final answer: Zero.

Question 11

Working: Centripetal acceleration a_c=v^2/r=12^2/6=144/6=24 m/s^2.

Final answer: 24 m/s^2.

Question 12

Working: For vertical fall from rest, h=(1/2)gt^2. Thus 20=5t^2, t^2=4 and t=2 s.

Final answer: 2 s.

Question 13

Working: For perpendicular components, R=sqrt(6^2+8^2)=sqrt(36+64)=sqrt100=10.

Final answer: 10.

Question 14

Working: 120 rpm=120/60=2 revolutions/s. Angular speed=2pi f=2pi(2)=4pi rad/s.

Final answer: 4pi rad/s.

Question 15

Working: $u_y=20\sin30=10$ m/s. Time $T=2u_y/g=20/10=2$ s. Maximum height $H=u_y^2/(2g)=100/20=5$ m.

Final answer: T=2 s; H=5 m.

Question 16

Working: Resultant $R=\sqrt{8^2+6^2}=\sqrt{100}=10$ N. Angle north of east $=\tan^{-1}(6/8)=36.9$ deg.

Final answer: 10 N at 36.9 north of east.

Question 17

Working: $a_c=v^2/r=10^2/25=100/25=4$ m/s^2 toward the centre.

Final answer: 4 m/s^2 inward.

Question 18

Working: Horizontal component $u_x=25(0.8)=20$ m/s; vertical component $u_y=25(0.6)=15$ m/s. Flight time $=2u_y/g=30/10=3$ s; range $=u_x t=20(3)=60$ m.

Final answer: 3 s; 60 m.

Question 19

Working: Vertical component $=20\sin30=10$ m/s. Time $=2(10)/10=2$ s. Height $=10^2/(2(10))=5$ m. Range $=20\cos30 \times 2=34.6$ m (equivalently $u^2\sin60/g$).

Final answer: 2 s; 5 m; 34.6 m.

Question 20

Working: Centripetal force $F=mv^2/r=5(4^2)/2=40$ N inward. The force is perpendicular to instantaneous displacement, so work $W=Fs\cos90$ deg $=0$ J.

Final answer: 40 N inward; centripetal force does zero work.